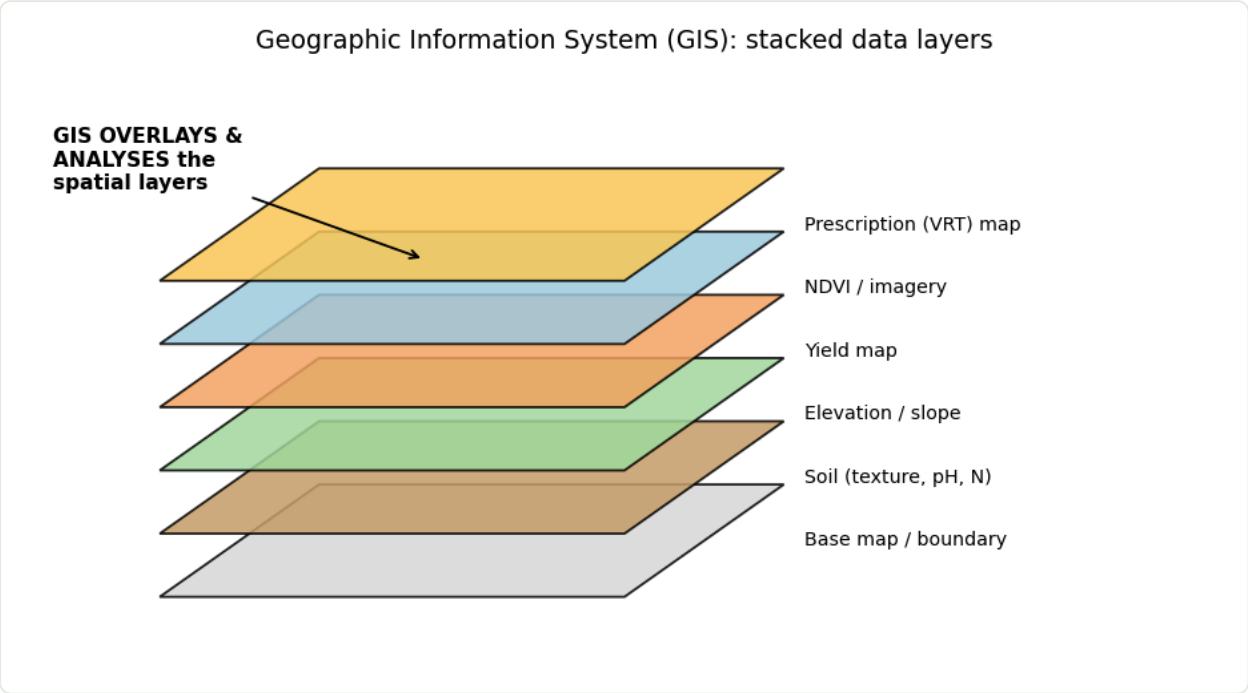
UNIT 10 · TOPIC 4

GIS — Functions and Use for Decision-Making

Syllabus topic: GIS: functions and use for decision-making

A **GIS (Geographic Information System)** is a computer system to CAPTURE, STORE, manage, ANALYSE and DISPLAY spatially referenced (geographic) data.



GIS — stacked data layers

- Information is held as **LAYERS** — soil, elevation, yield, imagery, boundaries — each tied to location, which can be OVERLAID and queried.
- **Data models:** VECTOR (points, lines, polygons — e.g., sampling points, field boundary) and RASTER (grid cells — e.g., satellite imagery, interpolated maps).

GIS function	Use in PA
Overlay analysis	Combine soil, yield and imagery layers
Interpolation (e.g., kriging)	Turn soil samples into a continuous surface
Buffering; proximity and network analysis	Distances from features, routing
Map production	Yield, soil and prescription maps

In PA, GIS integrates all the georeferenced data (GPS positions — Topic 3) and generates the management ZONES and PRESCRIPTION MAPS that drive variable-rate application (Topics 8–9).

Practice MCQs

Q36. A GIS is primarily used to:

(a) Drive the tractor
(b) Capture, store, analyse and display spatial data
(c) Measure temperature
(d) Weld metal

Q37. In GIS, points, lines and polygons are ___ data:

(a) Raster (b) Vector

(c) Analog (d) Thermal

Q38. Satellite imagery stored as grid cells is ___ data:

(a) Vector (b) Raster

(c) Tabular (d) Scalar

Q39. Building a continuous soil surface from scattered sample points uses:

(a) Buffering (b) Interpolation (e.g. kriging)

(c) Welding (d) Sampling only

Q40. The main output of GIS analysis that drives VRT is the:

(a) Yield only (b) Prescription (application) map

(c) Weather forecast (d) Fuel bill

Answers

Q36 — (b) Capture, store, analyse and display spatial data

Q37 — (b) Vector

Q38 — (b) Raster

Q39 — (b) Interpolation (e.g. kriging)

Q40 — (b) Prescription (application) map

Sources: Srinivasan — Handbook of Precision Agriculture; ICAR/NIIT PA modules; Lillesand & Kiefer — Remote Sensing and Image Interpretation; ISO 11783 (ISOBUS); remote-sensing and NDVI references (as given in the source notes, July 2026); NPTEL precision-agriculture lectures.